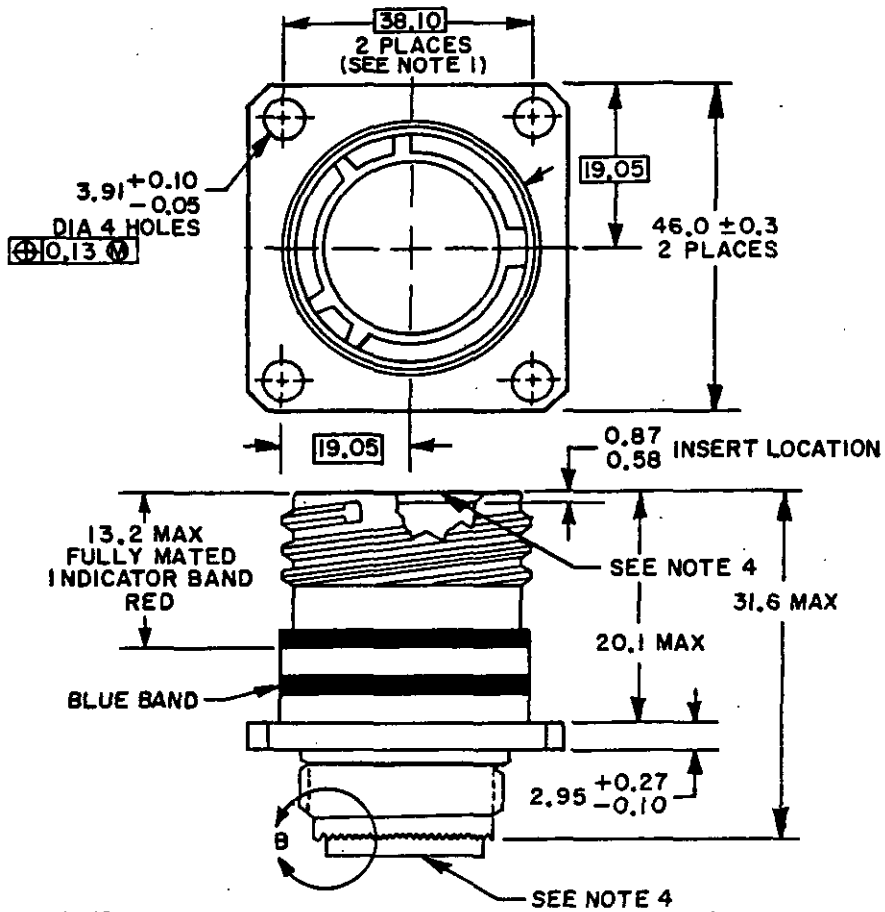
CONNECTOR, RECEPTACLE, ELECTRICAL, CIRCULAR, BREAKAWAY, WALL MOUNTING FLANGE, REMOVABLE CRIMP CONTACTS, SOCKETS, SERIES III, SHELL SIZE 25, METRIC

This specification sheet is approved for use within the Air Force by the 2750 Air Base Wing, Electronic Support Division (2750 ABW/ES), Gentile AFS, OH 45444-5400, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-38999.




| mm | Inches |
|---|---|
| 0.05 | .002 |
| 0.10 | .004 |
| 0.27 | .0106 |
| 0.3 | .01 |
| 0.58 | .023 |
| 0.87 | .0343 |
| 2.95 | .116 |
| 3.9 | .154 |
| 12.95 | .510 |
| 13.20 | .519 |
| 19.05 | .750 |
| 20.1 | .79 |
| 31.6 | 1.244 |
| 38.1 | 1.50 |
| 46.0 | 1.81 |

NOTES:
1. Dimensions are in millimeters.
2. Inch equivalents are given for general information only.
3. Front or rear panel mounting.
4. For details of connector front and rear configurations, see the procurement specification.

FIGURE 1. Receptacle class W.


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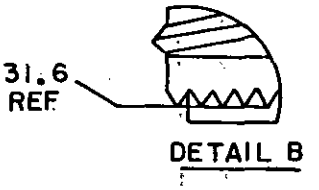



FIGURE 1. Receptacle class W - Continued.

REQUIREMENTS:

Dimensions and configurations: See figure 1.

Interface dimensions shall conform to MIL-C-38999.

This receptacle is designed for use with lanyard release plug and mates with DOD-C-38999/29, /30, and /31. It is intended to be used only on stores.

Insert arrangement: See MIL-STD-1560.

Separation force: The receptacle shell shall break into two parts when a separation force is applied at any angle within 15° of the normal axis by the mating plug assembly. Separation shall occur such that the plug assembly shall be pulled free of the receptacle insert and contacts shall remain secured in the mounted part.

QUALIFICATION:

Connectors shall meet the qualification requirements of MIL-C-38999, with the following exceptions: The mating plug to be used during tests shall be DOD-C-38999/31, type 2. For separation force tests, use DOD-C-38999/26.

Group I:

Durability: Wired connectors shall meet the durability requirements of MIL-C-38999, with the following exceptions:

The total cycles of mating and unmating shall be 500, in the following sequence: 200 cycles of normal mating and unmating, 50 cycles of normal mating with pull-separation unmating, 200 cycles of normal mating and unmating, and 50 cycles of normal mating with pull-separation unmating. The lanyard release velocity during the pull-separation unmating cycles shall be 9.15 meters per second.

Separation force: On the conclusion of the group I tests, all samples shall be subjected to an axial force applied by a mated plug assembly. The connector shall meet the requirements of separation force when the applied force is 1800 ±200 newtons.



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External bending moment: Wired connectors shall meet external bending moment requirements of MIL-C-38999 with the following exception: Size 25 loading shall be 28.3 newton-meters.

Group II:

Durability: Wired connectors shall meet the durability requirements of MIL-C-38999, with the following exceptions:

The total cycles of mating and unmating shall be 500, in the following sequence: 200 cycles of normal mating and unmating, 50 cycles of normal mating with pull-separation unmating, 200 cycles of normal mating and unmating, and 50 cycles of normal mating with pull-separation unmating. The lanyard release velocity during the pull-separation unmating cycles shall be 9.15 meters per second.

Vibration: Wired connectors shall meet the vibration requirements of MIL-C-38999 with the following exceptions:

Sine vibration: Connectors shall be subjected to the test specified in method 204, test condition G, MIL-STD-202.

Random vibration: Connectors shall be subjected to the test specified in method 2005 of MIL-STD-1344, test condition VI, letter J, ambient temperature. Duration shall be 8 hours in the longitudinal direction and in a perpendicular direction, for a total of 16 hours.

The QPL evaluating activity will define the accessory load and cable to be used in the random and sine vibration tests.

Separation force: On the conclusion of the group II tests, all samples shall be subjected to an axial force applied by a mated plug assembly. The connector shall meet the requirements of separation force when the applied force is 1800 ±200 newtons.

External bending moment: Wired connectors shall meet external bending moment requirements of MIL-C-38999 with the following exception: Size 25 loading shall be 28.3 newton-meters.

Part or Identifying Number (PIN) example: D38999/ 35 W 20 A

Military prefix

Specification sheet number

Class

Insert arrangement

Polarizing position
N is required for normal position

CONCLUDING MATERIAL

Custodian:
Air Force - 85

Review activities:
Air Force - 11, 15, 99

Preparing activity:
Air Force - 85

Agent:
DLA - ES

(Project 5935-F355-2)



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